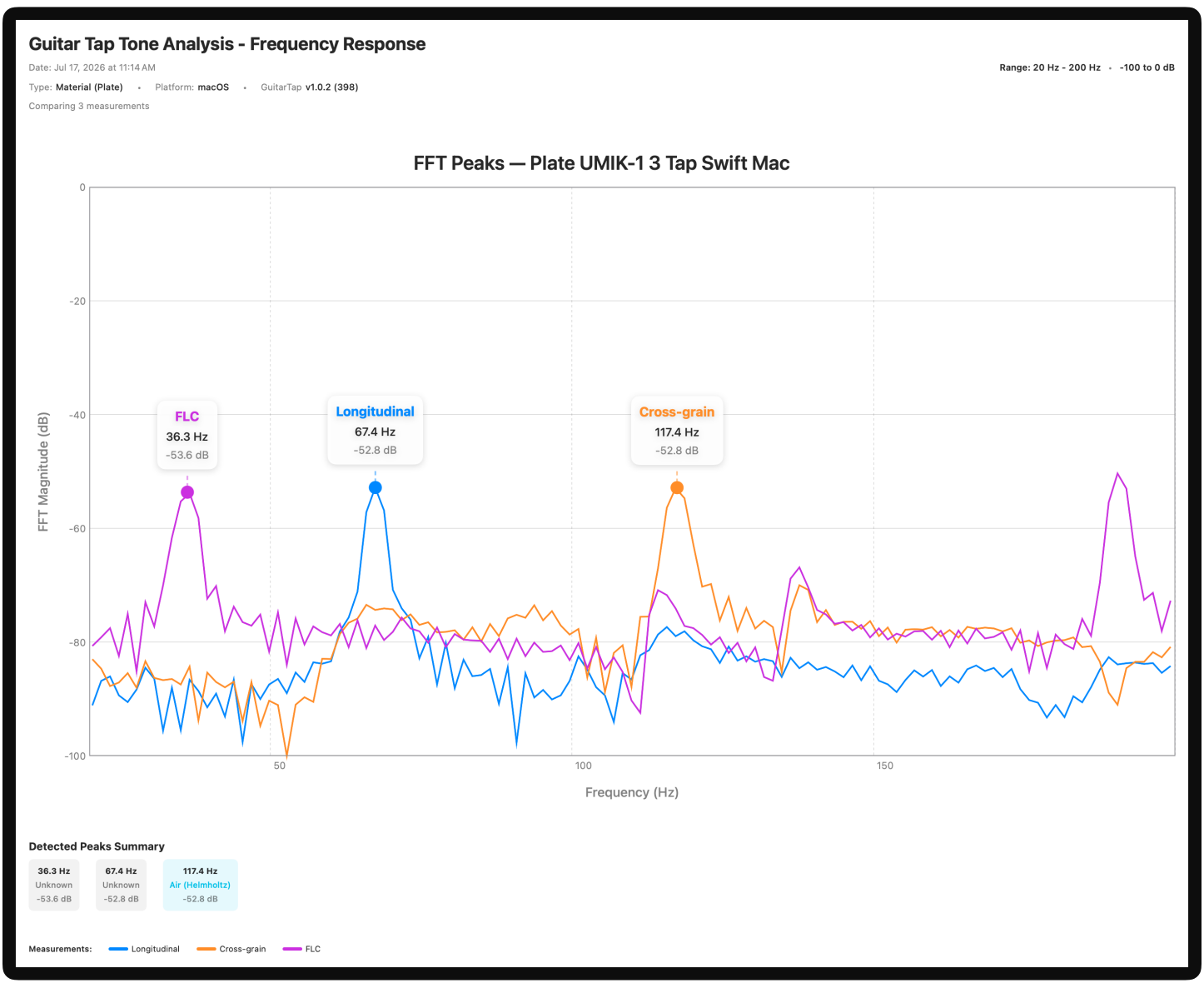
GuitarTap

Tap Tone Analysis Report

Jul 17, 2026 at 11:14 AM

Measurement Name: Plate UMIK-1 3 Tap Swift Mac
Type: Material (Plate)
Frequency Range: 20.0 Hz – 200.0 Hz
Microphone: Umik-1 Gain: 18dB · calibrated (7108913)

Frequency Spectrum



Detected Peaks

Frequency	Magnitude	Note	Q Factor	Role
36.3 Hz	-53.6 dB	D1	8.3	FLC (Diagonal)
67.4 Hz	-52.8 dB	C#2	23.0	Longitudinal (L)
117.4 Hz	-52.8 dB	A#2	26.7	Cross-grain (C)

Gore Target Thickness

2.79 mm

Body: 490 × 368 mm · f_vs = 60 (Classical Top)
GLC (Shear Modulus): 0.358 GPa

Plate Properties

Sample Dimensions

Length: **557.5 mm** Width: **220.5 mm** Thickness: **4.85 mm**
Mass: **208.0 g** Density: **0.349 g/cm³**

fL: **67.4 Hz** fC: **117.4 Hz** fLC: **36.3 Hz**

Speed of Sound (L): **4203 m/s**
Speed of Sound (C): **1145 m/s**
Young's Modulus (L): **6.16 GPa**
Young's Modulus (C): **0.46 GPa**
GLC (Shear Modulus): **0.358 GPa**

Specific Modulus (L): **17.7** (Fair)
Specific Modulus (C): **1.3** (Very Good)
Radiation Ratio (L): **12.0**
Radiation Ratio (C): **3.3**

Cross/Long Ratio: **0.074**
typical: 0.04–0.08

Long/Cross Ratio: **13.5**
typical: 12–25

Overall Quality: **Good**

Three-Tap Measurement Process:

1. **Longitudinal (L) Tap**
   Hold plate at 22% from one end along the length, near one long edge (not at the width node). Tap center.
2. **Cross-grain (C) Tap**
   Rotate 90°. Hold plate at 22% from one end along the width, near one short edge (not at the length node). Tap center.
3. **FLC (Diagonal) Tap**
   Hold plate at the midpoint of one long edge. Tap near the opposite corner (~22% from both the end and the side). Measures shear stiffness.

The strongest peak from each tap is auto-selected.